

OFFLOAD · V2 ROUTING

A PERMIT PDF INTO A ROUTABLE CORRIDOR.

Structure - first v2 extraction reads the corridor spine.

A self-healing routing tail keeps the truck on the permitted roads.

Proven on five real permits, flagship IN Baggett 0072232.

IN Baggett 0072232 [FULL]

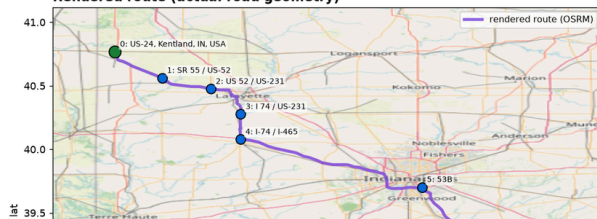
state=IN | origin boundary ok | dest boundary ok | corridors placed 16/16 | unplaced 0 | GEOMETRY ANOMALY

Turn-by-turn (Review > Route(s))

Via Route: 16 steps, 16 placed

1. Start on Origin: US-24,IL:Kentland at US-24, IL:Kentland
2. Turn right onto SR-55
3. Bear left onto US-52
4. Continue straight on US-231
5. Merge onto I-74
6. Merge onto I-465
7. Bear right onto I-65
8. Bear right onto US-31
9. Bear left onto SR-7
10. Continue straight on SR-3
11. Continue straight on SR-56
12. Merge onto I-65

Rendered route (actual road geometry)



THE PROBLEM

THE ENGINE SECOND-GUESSED A CORRECT READ

v2's LLM read the corridor spine correctly on every permit tested. The failures a customer saw were all downstream, in the deterministic routing engine.

What v2 extraction got right

The corridor spine, in order, on every permit: US-231, I-74, I-465, I-65 for the flagship. The permit is the law.

What the router did with it

Placed the "Merge onto I-74" waypoint on the far Cincinnati side of the concurrency, then swung the truck up the east I-465 beltway. A half-circle around Indianapolis, 315 miles.

The fix does not re-extract. It makes the router **honor** what the LLM read, and heals the placements that land on the wrong vertex.

STRUCTURE FIRST, THEN A SELF-HEALING TAIL

1. Structure-first extraction

Textextract structure, the RT detector stack, and an LLM route-expert fanout turn the permit into a corridor spine: an ordered list of roads and boundary terminals, every token copied verbatim from the source.

2. Deterministic routing

OSM road facts plus OSRM geometry place each corridor as a waypoint. Deterministic code is used only for data-facts (how OSM tags a road), never for reasoning.

3. The self-healing recovery tail (where this PR lives)

Resolver placement, boundary-terminal validation, and a geometry-anomaly detector that runs real OSRM road distance. When a waypoint lands on the wrong vertex, the healer re-snaps it onto the correct point of its own permitted road. Two new heals landed here.

PAIRWISE JUNCTION-DESTEAL

A "continue on ROAD_A" waypoint stole the ROAD_A x ROAD_B junction, evicting the "merge onto ROAD_B" waypoint to the wrong side of a metro concurrency.

The bug

IN Baggett: wp3 "Continue on US-231" sat AT the US-231 x I-74 junction, so wp4 "Merge onto I-74" was evicted to the SE I-465 beltway. A 5.97x wp4 to wp5 backtrack.

The heal

The merge waypoint snaps to the ROAD_A x ROAD_B entry junction, the continue waypoint slides upstream on ROAD_A. Accept only if the worst per-leg detour ratio strictly drops AND every permitted corridor is still traversed.

Live on IN Baggett: wp4 to wp5 detour **5.97x down to 1.67x**, all four corridors intact, 3 runs byte-identical.

LOOP / BELTWAY CANDIDATE SEARCH

A merge onto a loop road (a beltway) has junctions on two sides. A single deterministic snap can pick the side that sends the route the long way around.

The bug

IN Baggett wp5 "Merge onto I-465" swung the truck up the EAST beltway through Lawrence before coming back down. The swing is smooth, so the per-leg detour ratio misses it.

The heal

Enumerate every loop-road junction on both sides, OSRM-score the full sequence with the waypoint at each, move to the lowest-distance side. Accept only if it reduces the swing AND keeps every permitted corridor.

Live on IN Baggett: total **315.3 miles down to 286.7 miles**, all four corridors intact, 3 runs byte-identical.

IN BAGGETT 0072232: THE CLEAN CORRIDOR

Half-circle backtrack around Indianapolis to a clean US-231, I-74, I-465, I-65 corridor. 16 of 16 corridors placed, in order, both state-line terminals correct.

IN Baggett 0072232 [FULL]

state=IN | origin boundary ok | dest boundary ok | corridors placed 16/16 | unplaced 0 | GEOMETRY ANOMALY

Turn-by-turn (Review > Route(s))

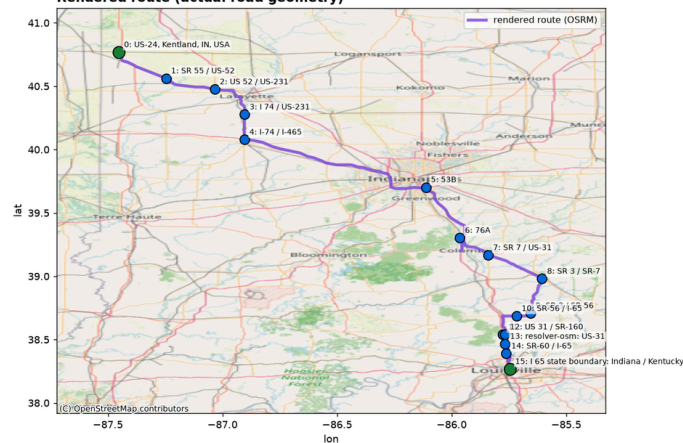
Via Route: 16 steps, 16 placed

1. Start on Origin: US-24,IL;Kentland at US-24, IL;Kentland
2. Turn right onto SR-55
3. Bear left onto US-52
4. Continue straight on US-231
5. Merge onto I-74
6. Merge onto I-465
7. Bear right onto I-65
8. Bear right onto US-31
9. Bear left onto SR-7
10. Continue straight on SR-3
11. Continue straight on SR-56
12. Merge onto I-65
13. Turn left onto SR-160
14. Turn right onto US-31
15. Turn right onto SR-60
16. Merge onto I-65 at the Kentucky state line

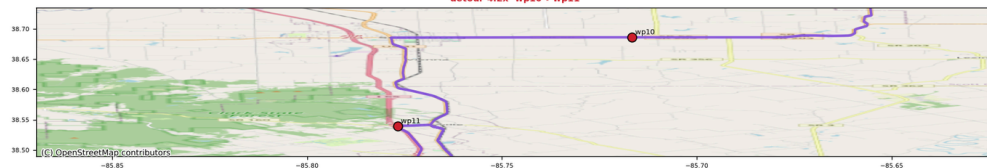
Geometry anomalies (from real OSRM road distance):

- detour wp10->wp11: 4.2x (72km road vs 17km straight)

Rendered route (actual road geometry)



detour 4.2x wp10->wp11



CORRECTION-CAPTURE: A DRAG TEACHES A RULE

The interactive proof tool lets a human drag a mis-placed waypoint. That approximate correction teaches a general rule, not a memorized coordinate.

The loop

1. A human drags the waypoint to roughly the right spot.
2. The rule is validated by geometry (does the OSRM detour drop, do the corridors hold), never by matching the dragged coordinate.
3. The rule is coded into the self-healer, re-run, and proven on 3 byte-identical runs.

Why it generalizes

The two heals above came out of this loop. Neither hard-codes Indianapolis Junction-desteal and loop candidate-search fire on any metro concurrency or beltway that matches the signature, and stay inert on the control cohort.

The permit is truth. Every heal is gated to keep the route on the permitted corridors, or it is rejected and left annotated for review.

END TO END

THE SAME FIX, DRIVEN THROUGH THE APP

Upload a permit, digitize it, and the corridor renders on the load's Routing & Permits tab. Multi-state loads stitch each state's permit into one official route sequence.

The screenshot displays the 'offload' app interface for a user named 'Schaefer Freight Brokers'. The top navigation bar includes a search field 'Jump to project...' and a user profile icon. The left sidebar contains navigation options: Projects, Loads, Assets, Contacts, and Sharing. The main content area is titled 'PROJECT P-501-2' and shows a load 'L-501-1' in 'Planning' status. The load details include the route 'Chicago, Illinois to Atlanta, Georgia' and pickup/drop times. The 'Routing & Permits' tab is active, showing a map of the route from Chicago to Atlanta. The map displays the route passing through Illinois, Indiana, Kentucky, Tennessee, and Georgia. The left sidebar under 'Routing & Permits' shows 'Total Distance: 1753.83 miles', 'Max Escorts: 0', and 'Total Permits: 4'. It also displays 'State Permit Progress' for IL, KY, TN, GA, and IN. The 'Routes' section lists the 'OFFICIAL ROUTE SEQUENCE' with stops in Chicago, Illinois, and Atlanta, Georgia, and a list of permits for each state: IL - #ST2026692843, KY - #2026-546-47139, TN - #L266784167266, and GA - #L263562877678. The 'NOT IN OFFICIAL SEQUENCE' section shows 'No routes'.

offload / Schaefer Freight Brokers

PROJECT P-501-2

L-501-1 Planning Switch

Chicago, Illinois to Atlanta, Georgia

PICKUP Aug 2, 10:54 AM EDT to DROP Aug 11, 1:09 PM EDT

Details Dispatch Routing & Permits Documents People History

Routing & Permits Digitize Permits

Total Distance: 1753.83 miles

Max Escorts: 0

Total Permits: 4

State Permit Progress

IL KY TN GA IN

Routes Permits

+ Add Planned Route

OFFICIAL ROUTE SEQUENCE

- 2016 Steuber Circle, Chicago, Illinois 60601
- Planned Route
- IL - #ST2026692843 Approved
- KY - #2026-546-47139 Approved
- TN - #L266784167266 Approved
- GA - #L263562877678 Approved
- 6188 St George's Road, Atlanta, Georgia 30301

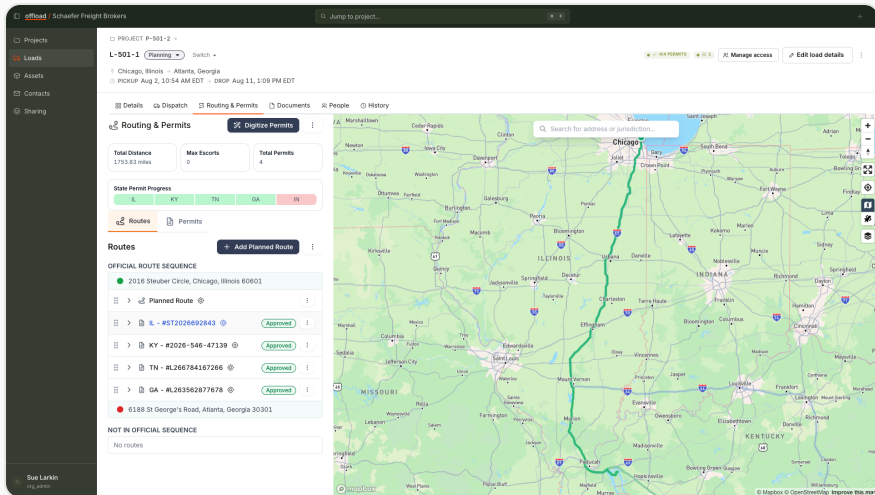
NOT IN OFFICIAL SEQUENCE

No routes

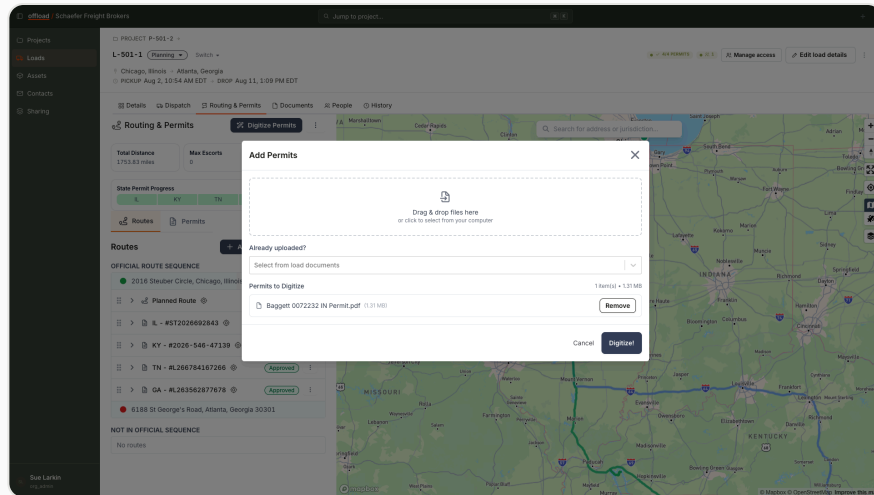
Sue Larkin org_admin

END TO END

ONE CLICK PER CORRIDOR, ONE DIGITIZE PER PERMIT



Selecting the IL permit highlights just its corridor (I- 57 down through Illinois) on the map.



The digitize dialog: drop a permit PDF, click Digitize, and the v2 pipeline runs it into corridors.

IL BAGGETT: A BOUNDARY-TO-BOUNDARY CORRIDOR

6 of 6 corridors placed, WI state line to IN state line, zero geometry anomalies. Every leg tracks its straight line.

IL Baggett 0072232 [FULL]

state=IL | origin boundary ok | dest boundary ok | corridors placed 6/6 | unplaced 0

Turn-by-turn (Review > Route(s))

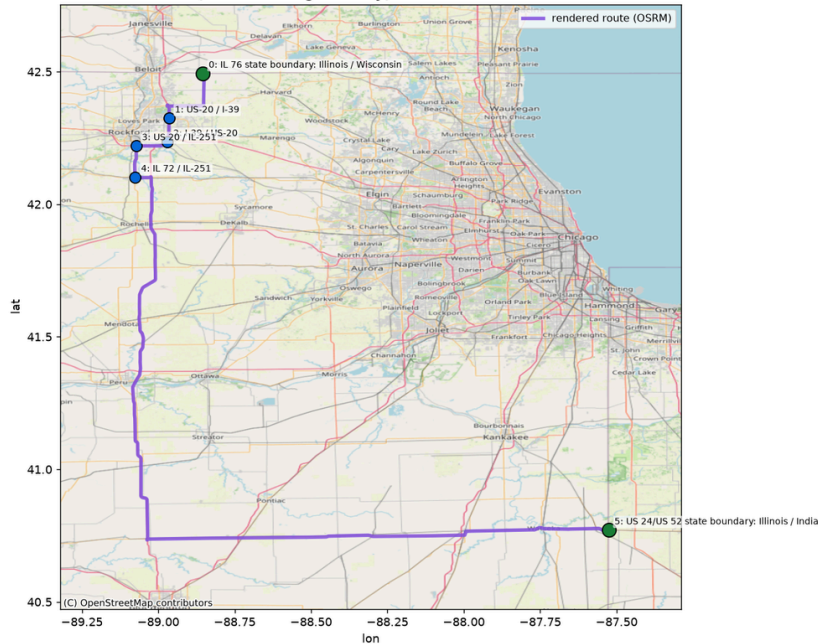
Via Route: 6 steps, 6 placed

1. Border Start: Wisconsin - IL-76 at the Wisconsin state line
2. Continue on Make sharp right on US-20
3. Bear right onto ramp and go on US-20 W/1-39 S /US-51 S toward FREEPORT / BLOOMINGTON
4. At fork keep left on US-20
5. Take ramp on the right to IL-251 S toward ROCHELLE
6. Turn left on IL-72 at the Indiana state line

Geometry anomalies (from real OSRM road distance):

- none: every leg tracks its straight line, no loops or U-turns

Rendered route (actual road geometry)



WI BAGGETT: THREE CORRIDORS, NO ANOMALIES

3 of 3 corridors placed, IL/WI boundaries correct, zero geometry anomalies.

WI Baggett 0072232 [FULL]

state=WI | origin boundary ok | dest boundary ok | corridors placed 3/3 | unplaced 0

Turn-by-turn (Review > Route(s))

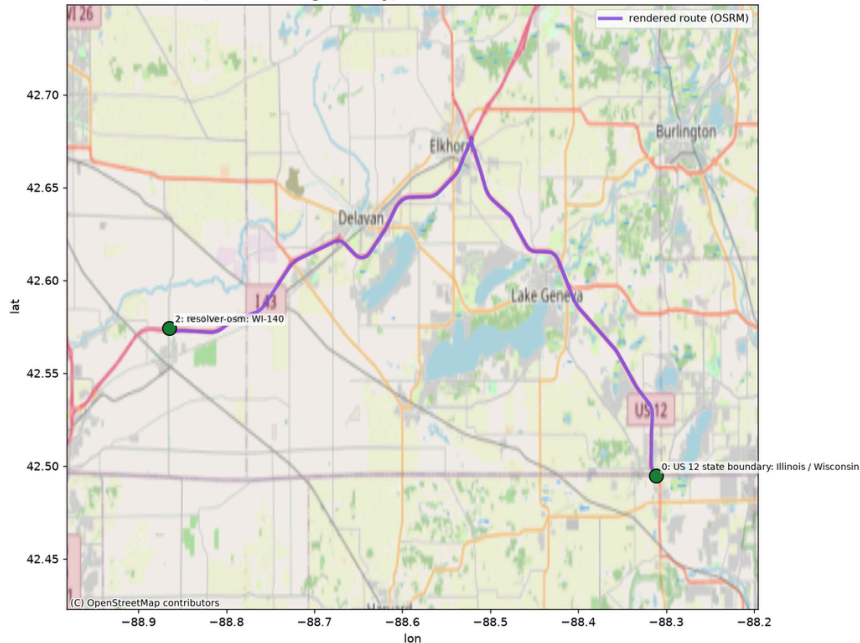
Via Route: 3 steps, 3 placed

1. START ON US-12 WB AT MP WALWORTH 340.35(STATE BORDER OF ILLINOIS)
2. Continue on I-43 SB
3. End on WI-140 SB at the Illinois state line

Geometry anomalies (from real OSRM road distance):

- none: every leg tracks its straight line, no loops or U-turns

Rendered route (actual road geometry)



TN BAGGETT: THE NASHVILLE I-840 BYPASS

3 of 3 corridors placed, I-65 to I-840 (the southern bypass) back to I-65. The anomaly detector flags the tight interchange pair honestly rather than hiding it.

TN Baggett 0072232 [FULL]

state=Tennessee | origin boundary ok | dest boundary ok | corridors placed 3/3 | unplaced 0 | GEOMETRY ANOMALY

Turn-by-turn (Review > Route(s))

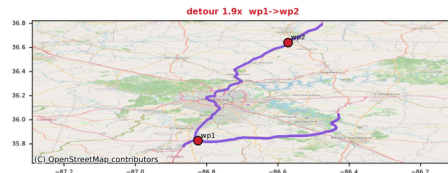
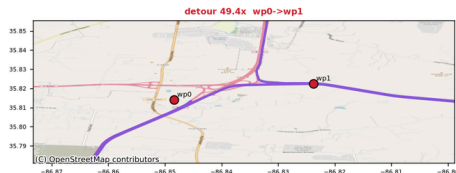
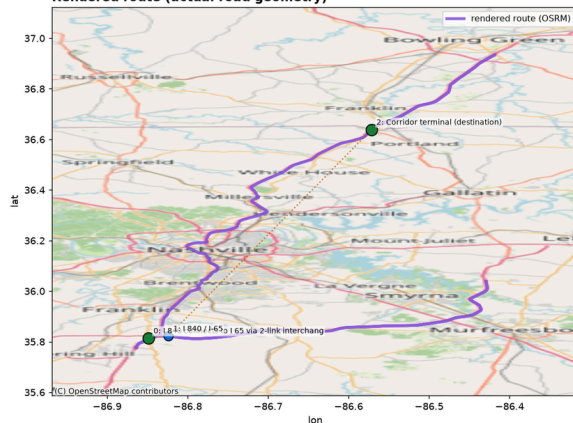
Via Route: 3 steps, 3 placed

1. Start on I 65
2. Continue on I 840
3. End on I 65

Geometry anomalies (from real OSRM road distance):

- detour wp0->wp1: 49.4x (119km road vs 2km straight)
- detour wp1->wp2: 1.9x (174km road vs 94km straight)
- (context: whole route is 3.1x the origin-to-dest straight line)

Rendered route (actual road geometry)



CO 24SB0009511: A CLEAN SPINE, METRO WOBBLE

The interstate spine from Denver to the Nebraska line renders clean. The dense Denver-metro merge and exit waypoints wobble, and the destination boundary missed.

CO 24SB0009511 [PARTIAL]

state=CO | origin boundary ok | dest boundary MISS | corridors placed 5/4 | unplaced 0 | GEOMETRY ANOMALY

Turn-by-turn (Review > Route(s))

Via Route: 4 steps, 5 placed

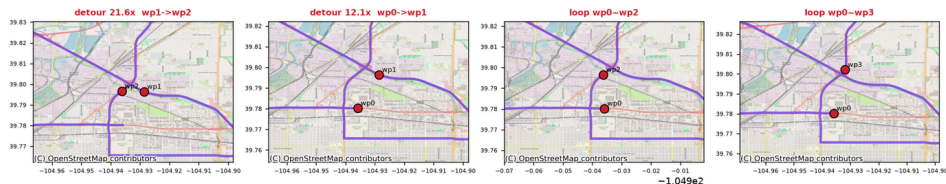
1. Continue straight on I-70 [US-36]
2. Take Exit 279A toward I-270 W/FORT COLLINS/BOULDER
3. Merge onto US-85 [US-6], [CO-2], [VASQUEZ BOULEVARD]
4. Continue straight on I-76 [US-6], [US-85] at the Nebraska state line

Rendered route (actual road geometry)



Geometry anomalies (from real OSRM road distance):

- detour wp1->wp2: 21.6x (14km road vs 1km straight)
- detour wp0->wp1: 12.1x (23km road vs 2km straight)
- loop/track: wp0 and wp2 only 1.8km apart
- loop/track: wp0 and wp3 only 2.5km apart
- loop/track: wp1 and wp3 only 0.7km apart



HONEST

WHAT IS STILL OPEN

The KY-end detour on the flagship

IN Baggett wp10 to wp11 (US-31 to I-65 near the Kentucky line) is a 4.2x detour, a separate defect from the two heals here. It is byte-identical to base, not worsened, and out of scope for this PR.

No automated route-correctness gate yet

Every offline gate here grades extraction, not whether the rendered route matches the permit. The cohort validation on these slides is manual. A route-vs-Via-Route gate is being built separately.

The Via Route these renders score against is itself LLM-extracted, not ground truth. Driven-GPS recall is the independent ruler.

ALL OF THIS IS V2-GATED

v1 byte-identical

Every heal is behind the v2 seam. v1's route output does not move.

Gold gate every commit

The offline gold, invariants, and synthetic-regression gates prove v1 is unchanged on each commit.

The permit is truth

No heal reroutes off a permitted corridor. It corrects a placement or it is rejected.

Merging this ships the improvements behind the master flag. It does not activate v2 in prod. Flipping the flag stays a separate, later decision gated on broader v2-beats-v1 breadth.